CAP 3330

Module 4: Linear Regression

Topics

- Correlation
- Simple linear regression
- The estimated regression equation

Suggested reading. Chapter 3 from the book “An Introduction to Statistical Learning: with applications in R” (**Free version online. Google it if interested**)

Linear regression

Goal: Predict a quantitative **dependent/outcome/target/response** variable (**Y**) based on one or more **predictor/ independent/ input/ feature** variables (**Xs**).

One ***assumption*** in linear regression is that Y and X (or Xs) are linearly related (= correlated). Two ways of ***checking if this assumption*** is satisfied are:

a) Compute **r**, the sample correlation coefficient between Y and X.

b) Do a scatterplot of Y vs X.

Correlation coefficient

- The sample correlation coefficient, **r**, measures the degree of linear relationship between two variables.

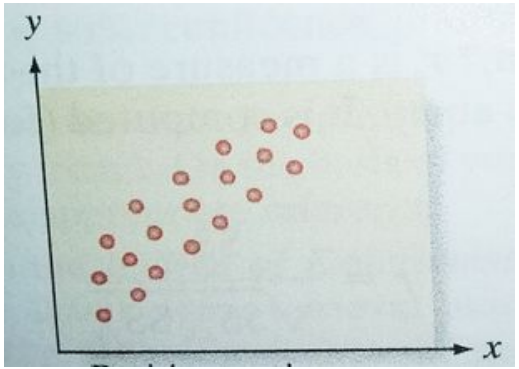
r formula

$$\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

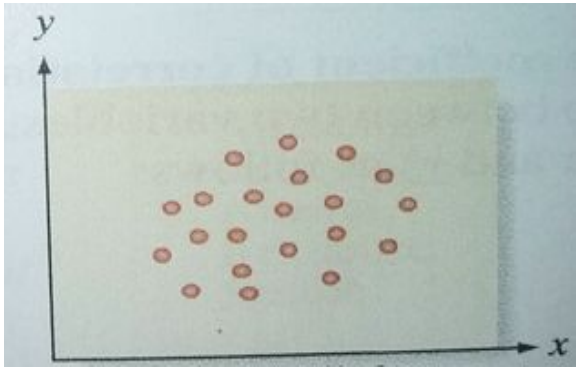
$-1 \leq r \leq 1$

- The closer r is to 1, the clearer the sign of strong positive correlation.
- The closer r is to 0, the clearer the sign of no correlation.
- The closer r is to -1, the clearer the sign of strong negative correlation.

Correlation (graphically): The scatterplot



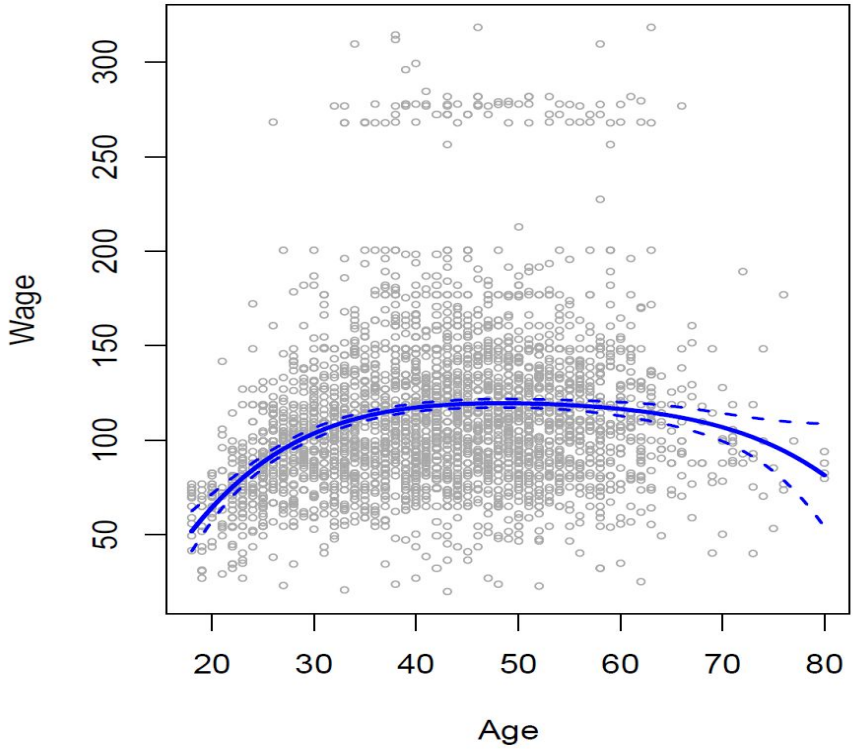
Graph shows a strong positive correlation between Y and X (r close to 1)



r close to 0. Why?

Because the variables do not show a linear (or any kind of) relationship

More on the scatterplot



Relationship and causation

Now, get ready for the most repeated sentence in Statistics …

Correlation does not imply causation

A predictor X can be highly correlated with an outcome Y, but that **does not mean** that **we can be certain** that high values of X is what causes high value of Y.

Example 1: High positive correlation between X (monthly sales of beer in a store close to the beach) and Y(the number of monthly shark attacks).

Example 2: The ANOVA example about the effect of feed supplement (X) on the chickens' weight (Y). We showed that different diets are linked to different average weights. Can we claim, for certain, that some diet cause weight gain?

An observed correlation RARELY implies causation because the observed correlation is USUALLY explained by ***other confounding predictor variables***.

The case for causation is only defensible if the analyst can carefully isolate the effect of the supposed cause (= the predictor under study) by removing the effect of all other confounding variables.

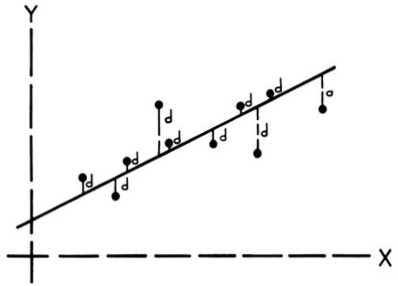
Go to R Studio to practice.

Estimation of the linear equation

- We take a sample of **n** observations and measure the values of X and Y on each observation. **For example**, we take a sample of **n** people and measure height (X) and weight (Y) for each of them.

- We apply a method called **Least Squares Method** to estimate the values of Bo and B1.

Estimation of the equation: **Least squares method**



The least squares line minimizes the sum of the squared distances.

Given the cloud of points representing the n sample observations, the least squares method **finds the equation** that best fits these points. The Least Squares method finds the value of the Y intercept and slope of the best equation.

Estimation of the linear equation

Estimated regression equation: Y hat

Y= Y hat + e

Y = bo + b1 * X + e = b1 * X + bo + e

The **Least Square Method**, based on mathematical optimization, tells us that we should use the following formulas to obtain b_o and b_1:

$$b_1 = r\frac{s_y}{s_x}$$

$$b_0 = \overline{y} - b_1\overline{x}$$

Estimated regression equation: Y hat

When we evaluate Y hat for a given value of X, the result is a predicted (estimated) value of Y.

Example: Comparing predicted Y VS real Y.

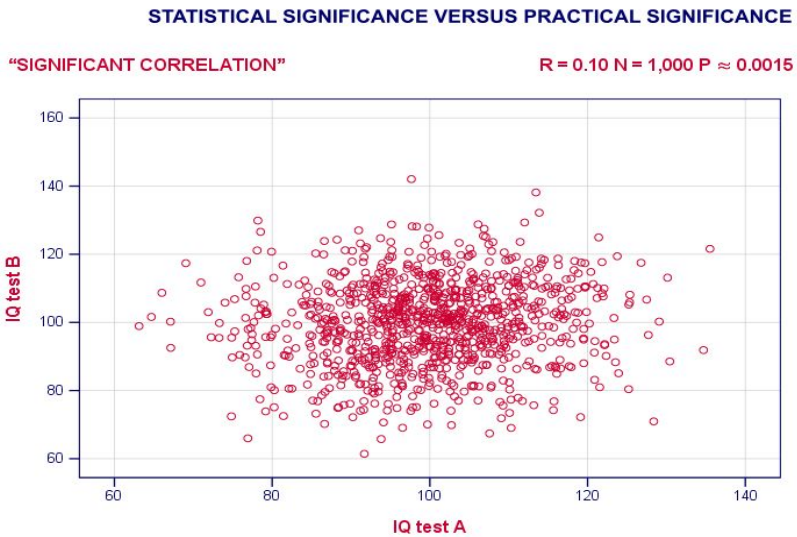
The next table shows the results of the Yearly Rainfall in San Francisco.

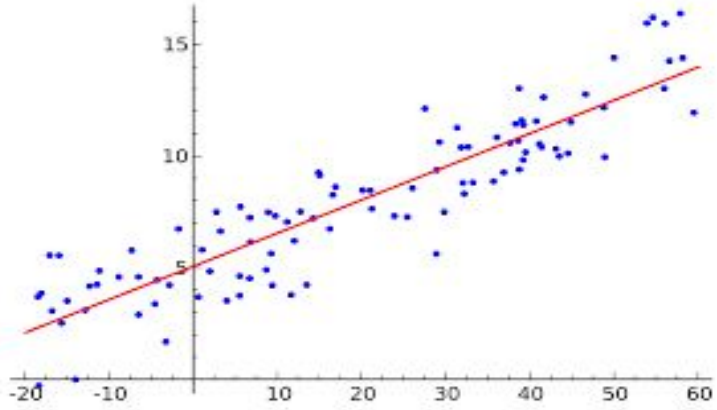
Year	Rain Days (x)	Observed Rainfall (y)	Predicted Rainfall ($\hat{y}$)
1997	119	47.22	43.7723
2005	104	34.43	37.1318
1982	100	38.17	35.3610
1994	100	34.02	35.361
1968	93	25.09	32.2621
1960	68	13.87	21.1946
1989	49	14.32	12.7833
1975	47	7.97	11.8979
1976	41	11.06	9.2417
...	...	...	...

LS method --------> $\hat{y} = 0.4427\ x - 8.909$ **------>**

Comparing the observed values of Y (third column) with the predicted values of Y (column on the right) gives us an indication of how good the equation is.

Statistical VS Practical Significance of the Regression Equation





Steps to evaluate the quality of **Y hat** in simple linear regression

1- **Statistical significance** (assessment of the statistical significance of equation)

- Does the mean of Y change significantly as the values of X change? When the values of X increase, does the change that happen in Y is statistically significant or just a random change?
- **How to evaluate it?** t test to see if **β1=0 or ≠ 0**. β1=0 (the slope of the equation equal to zero) means that the true line connecting Y and X is horizontal (X is not associated with Y).

2- **Practical significance** (assessment of the equation fit)

- How well does the equation predict the values of Y? Are predictions close to the actual values of Y?
- **How to evaluate it?** Coefficient of determination (R squared), Residual Standard Error (RSE)

Assessing the **statistical significance** of the equation (READ INDEPENDENTLY)

Test for significance of β_1

Hypotheses

Ho: $\beta_1=0$ (the mean of Y does not change significantly as X changes) (there isn't a statistically significant relationship between Y and X)

Ha: $\beta_1 \neq 0$ (the mean of Y changes significantly as X changes) (there is a statistically significant relationship between Y and X)

Test statistic

$$t = \frac{b_1}{\text{Std dev of } b_1}$$ With DF= n-2

Assessing **the fit (practical significance)** of Y hat

- R^2 (Coefficient of determination).
- R^2 = (SST - SSE)/ SST = 1 -(SSE/ SST)

SST: Sum of Square Total (variability of the Y values around **Y bar** (the sample mean of Y))

$$\sum_{i=1}^{n}(y_i - \bar{y})^2$$

SSE: Sum of Square of Error (variability of the Y values around **Y hat** (the regression equation))

$$\sum_{i=1}^{n}(y_i - \hat{y_i})^2$$

Assessing **the fit (practical significance)** of Y hat

- R^2 is the **percent reduction in variability (= error)** that we get **when we use the regression equation** instead of **the sample mean of Y** to make predictions of Y.

- **Despite its popularity,** it provides only limited information. It tells us how much better the regression equation is, compared to simply using the sample mean of the target variable, when predicting the target's values.

- $0 <= R^2 <= 1$. Larger values better. No straightforward criterion to establish a threshold.

Assessing **the fit (practical significance)** of Y hat

- **The Residual Standard Error** (RSE)

$$RSE = \sqrt{MSE}$$

where, **MSE** is the Mean Squared Error and ***measures the variance around the equation***.

MSE = SSE / (n-p-1)

where **p** is the number of predictors

- RSE is an estimate of the standard deviation of the data points around the equation.

Assessing **the fit (practical significance)** of Y hat

- **Roughly speaking**, RSE is an estimation of the average amount that the actual values of the target deviate from the regression line.
- How good (low) the RSE is ***depends on context.*** The smaller, the better.
- Since how good (low) RSE is depends on context, we can somewhat contextualize RSE by comparing it to the average value of Y. That's what the coefficient of variation does!

Coefficient of variation = *RSE / sample mean of the target*

This coefficient is a %. The closer to 0%, the lower (better) we can claim that RSE is.

Go to R Studio to practice.

Validity assumptions in linear regression

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- The **random error component** of this equation, **e**, is a random variable that should **satisfy certain conditions.** Otherwise, the *hypothesis test to assess statistical significance*, *the meaning of RSE*, and other analyses **may NOT be valid.**

- What are these conditions?

Assumptions for the validity of the linear regression analysis

Assumption 1: The relationship between Y and X is linear.

Assumption 2: The values of ε associated to any two observed values of Y are independent. That is, the value of ε associated with one value of Y has no effect on any of the values of ε associated with any other Y values.

Assumptions for the validity of the linear regression analysis

Assumption 3: The probability distribution of ε is Normal.

Assumption 4: The variability of ε is the same for all the values of X.



CAP 3330

Module 4: Linear Regression